

The Problem, and What We Built

What SIH26066 actually asks for, what OceanEmbed produces in answer, and who that answer is for.

PROBLEM STATEMENT	SPONSOR	REGION	GRID
SIH26066	Ministry of Earth Sciences / INCOIS	North Indian Ocean, 5-30 N / 45-105 E	0.25 deg - 100 x 240 = 24,000 cells

The ocean is opaque. Satellites, for all their power, only ever see its skin - the top millimetre or so of the surface. Almost everything that matters to a fisherman, a cyclone forecaster, a naval officer or a climate scientist is happening below that skin, where no satellite can look. SIH26066 asks us to close that gap.

1. What the problem statement asks for

In its own terms, SIH26066 asks for a system that reconstructs **depth-wise subsurface ocean temperature** from **daily surface satellite observations**, at **0.25 degree resolution** over the **North Indian Ocean**, at **15 standard depths from the surface to 1000 metres** - evaluated by RMSE, correlation and bias, using the GLORYS reanalysis as a training target and gridded Argo float data as independent validation.

Unpacked into plain requirements, that is seven distinct obligations, and a solution has to satisfy all of them at once rather than picking the convenient ones:

- **Inputs must be satellite observations** - sea-surface temperature, salinity, height, and surface currents and winds. Not reanalysis. Not model output. Observations.
- **Output must be a full profile** - 15 fixed standard depths, not a single number and not a smooth curve fitted afterwards.
- **Resolution must be 0.25 degrees** - roughly 25 kilometres, fine enough to resolve eddies.
- **Cadence must be daily** - because a monthly average erases cyclones, eddies and fronts, which are the events anybody actually needs.
- **The method must involve deep learning**, producing a compact satellite embedding that the reconstruction reads.
- **Evaluation must be quantitative** - RMSE, correlation and bias, against observations the model never saw.
- **It must work over the Bay of Bengal and the Arabian Sea**, which behave very differently from one another.

2. Why this problem exists at all

There are two ways to know the temperature a kilometre below the sea surface today, and both of them fall short.

The existing option	What it gives you	Where it fails
Argo floats - about 4,000 robotic instruments drifting worldwide, diving and surfacing on a cycle	Genuine, high-quality measurements of the real water column.	Sparse and slow. A float reports from wherever it happens to have drifted, roughly every 5 to 10 days. Over an entire basin, on any given day, almost every location has no float anywhere near it.
Ocean reanalysis - a physics model constrained by every observation available	A complete, physically consistent picture everywhere.	Expensive and delayed. It requires supercomputing, it lags real time, and it is itself an estimate - our own measurements show it carries around 1.1 degC of error at the thermocline.
Satellites	Complete coverage, every day, already paid for and already in orbit.	They only see the surface. Nothing an imaging satellite measures comes from below the skin of the ocean.

So the question SIH26066 poses is genuinely a good one: **can the daily, complete surface picture that satellites already provide be turned into the depth-resolved picture that only sparse floats and expensive models can currently give?** The physical basis for hoping so is real - the surface and the subsurface are dynamically coupled, and sea-surface height in particular is an integrated signal of what the whole water column is doing. The engineering question is how far that coupling can be pushed, and how honestly the remaining error can be reported.

THE ONE-SENTENCE VERSION OF THIS PROJECT

Argo floats answer 'what is the temperature at 500 metres' only where a float happens to be, every five to ten days. **OceanEmbed answers it at 24,000 locations, every single day, from satellites that are already in orbit** - and it reports how much to trust each answer.

3. What we actually produce

OceanEmbed is a complete system, not a single model. At its centre is **TS-Cast-NIO**: a 3-D convolutional encoder that compresses eleven days of seven satellite channels into a 128-number description of the surface state, feeding a compact head that turns those 128 numbers into fifteen depths - and which emits an uncertainty alongside every temperature it predicts.

Around that model sit seventeen working surfaces - each one a running dashboard or service on its own port - covering validation, physics, events, export, acoustics, cyclone analysis, robustness testing and observation planning. Each of the following seventeen notes covers exactly one of them.

What SIH26066 asks	What OceanEmbed delivers
Satellite inputs	7 channels harmonised from 5 distinct satellite products - OSTIA sea-surface temperature, a multi-mission salinity blend, DUACS sea-surface height, GLOBCURRENT surface currents and a CMEMS wind product. Verified by a 44-check bundle verifier that includes a negative test which deliberately injects reanalysis and confirms the check catches it.
15 standard depths	0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700 and 1000 metres - frozen as a constant in the code, imported everywhere, never retyped.
0.25 degrees, daily	100 x 240 = 24,000 cells, 388 consecutive daily steps from 1 June 2025 to 23 June 2026, with zero gaps .
Deep-learning embedding	A 3-D CNN producing a 128-dimensional latent , chosen over four competing architectures in a controlled bake-off where the prediction head was held identical so the encoder was the only variable.
RMSE, correlation, bias	All three at all 15 depths , against 962 independent Argo profiles and 12,736 depth comparisons.
Both basins	Scored separately per basin across three random seeds, with an Arabian Sea penalty reported as unexplained rather than smoothed over.

The headline result, on floats the model has never seen [VERIFIED]:

RMSE	Correlation	Bias	Skill over climatology	Independent profiles
0.9006 degC	0.8809	+0.1066 degC	+0.2400	962 (n = 12,736)

And the system is audited against the problem statement clause by clause, by a script that opens an artifact for every row rather than trusting a hand-written table: **16 requirements PASS, 0 FAIL, 1 BLOCKED**. The blocked one is independent validation against the INCOIS gridded Argo product specifically - their catalogue is reachable and carries exactly the right dataset, but their data-serving backend returns errors and empty responses. We validated against 962 independent float profiles from another source instead, documented the deviation, and call it BLOCKED rather than quietly counting it either way.

BLOCKED IS A REAL ANSWER, NOT A SOFTER WORD FOR FAIL

It means the requirement is understood, our side of the work is done, and an external dependency is unavailable. We wrote no downloader for it, because *a downloader that has never once retrieved a byte is not evidence of anything*. This must be stated to a jury as exactly that.

4. The one decision that defines this project

This is the most important thing on any of these nineteen pages, and it should be said out loud early in any presentation.

Configuration	RMSE vs Argo	Input source	Status
Stage 1, satellite inputs	0.9006 degC	OSTIA / DUACS / salinity blend / GLOBCURRENT / wind	SHIPPED - the deliverable
Stage 1, reanalysis inputs	0.8743 degC	reanalysis	comparator only
Stage 2, reanalysis inputs	0.8548 degC (unmasked_v1 protocol, n 12,829 - not directly comparable)	reanalysis	comparator only - the best number in the project, and NOT our result
Stage 2, satellite inputs	0.8854 / 0.9095 / 0.9158 across three seeds	satellite	not promoted, not frozen - the sign does not hold

Our most accurate configuration scores **0.8548 degC**. We do not ship it and we do not quote it as our result, because its inputs are reanalysis and the problem statement asks for satellite observations. Presenting it would be presenting a reanalysis-fed model as satisfying a satellite-input requirement. The freeze script **asserts** that the shipped artifact's input source reads *satellite*, so this cannot silently regress even if somebody wanted it to.

Against a reanalysis-fed comparator, real satellite observations read +0.026, +0.027 and -0.003 degC across three seeds on identical points - a mean of +0.017 whose sign does not hold. By our own three-seed rule that is not an established cost: the two inputs are within seed noise, and the satellite model retains about 94% of the comparator's skill.

The framing to use in front of a jury.

5. The impact this creates

The change OceanEmbed makes is not that subsurface temperature becomes knowable - it already is, from floats and reanalysis. The change is in **coverage, cadence and cost**. The same quantity becomes available everywhere in the basin, every day, from data that has already been paid for and is already being collected.

Domain	What changes on the ground
Cyclone forecasting	Tropical Cyclone Heat Potential - the subsurface heat that decides whether a storm intensifies - becomes available daily across the whole basin rather than only where a float surfaced. The Bay of Bengal produces a modest share of the world's cyclones and a very large share of its cyclone deaths, because the storms meet a low-lying, densely populated coast. Intensification lead time converts directly into evacuation lead time.
Fisheries and coastal livelihoods	Thermocline depth, fronts and upwelling zones govern where fish concentrate. Daily maps of them reduce the fuel and the time small-boat fishers spend searching - and reduce the risk they take in doing so.
Maritime operations and security	Sound speed decides how far a sonar hears and where its signal bends. Producing it daily from satellites rather than from survey vessels lowers the cost of maritime domain awareness and search-and-rescue capability across a very large ocean.
Climate monitoring	Ocean heat content is the single most direct measure of how much heat the planet is absorbing. Producing it daily over a basin with historically sparse observation adds to the record where the record is thinnest.
Research capacity	A reproducible, frozen, independently-validated reconstruction over a region where such products are scarce - exported as standard NetCDF and served over a documented API, so other groups can build on it without asking permission or re-implementing anything.

6. Who this is for, and why it matters to them

Who	Why it is important to them specifically
INCOIS and the Ministry of Earth Sciences	The sponsor. They operate India's ocean observation and services mandate. This gives them a daily basin-wide subsurface product from satellites they already receive - no new instrument, no new deployment, no new launch.
IMD and cyclone forecasters	The subsurface heat field is a direct input to intensity forecasting, and it is currently the part they have least of.
Coastal district administrations	Evacuation timing is the decision that saves the most lives in a cyclone, and it depends on how confident the intensity forecast is.
Fishing communities across India, Sri Lanka and the Gulf	Small operators who will never own an instrument gain access to information that previously required a research vessel.
The Indian Navy and Coast Guard	Acoustic conditions across the basin, daily, without a survey vessel.
Ocean and climate researchers	An open, provenance-carrying dataset over an under-observed basin, with the model's own error honestly characterised so it can be used correctly.
Students and educators	A rotatable, clickable, real-data picture of a layered ocean - built from genuine reconstruction rather than illustration.

7. How to see the whole system in ten minutes

Seventeen surfaces is a lot to demonstrate. This running order tells the strongest story, and each note in this pack contains the full instructions for its own feature.

- 1 Feature 7 - the model itself (port 8507).** Show the provenance block and the headline numbers. State the satellite-versus-reanalysis decision immediately.
- 2 Feature 12 - the click map (8512).** Hand it to the jury. Let them click. Then click on land and show the refusal.
- 3 Feature 8 - the Argo overlay (8508).** Let them choose a point, and check the model against a real float in front of them.

- 4 **Feature 3 - the Validation Lab (8503).** Show the worst numbers on purpose: the inherited error, the mixed layer that is genuinely ours, the depth where climatology wins.
- 5 **Feature 9 - cyclone heat (8509).** This is where the project becomes nationally relevant rather than merely accurate.
- 6 **Feature 17 - the cyclone case study (8517).** Close here. The model reproduces a cold wake nobody taught it, and the naive measurement that would have found nothing is shown beside it.

IF YOU ONLY HAVE THREE MINUTES

Feature 7 for the result, Feature 12 to let them test it, and Feature 17 for the finding that no error figure can substitute for.

8. What we do NOT claim

This belongs in the opening note rather than buried at the end, because a jury that hears the boundaries early trusts everything that follows.

- **We do not claim a novel reconstruction method.** AI reconstruction of subsurface temperature from surface data is established prior art, and our own literature matrix names the papers.
- **We do not claim novelty for uncertainty-aware reconstruction.** The paper we re-implemented does it, and does it better than we do.
- **We do not claim novelty for observation targeting.** Our own review found it is a formally-optimised research area; ours is a simple heuristic and we say so on the page itself.
- **We do not claim to be first in this basin.** Global methods already include the North Indian Ocean.
- **We never say the model tells anyone where to deploy floats.** The sanctioned wording is stored in the code and a test fails if either the module or the page drifts into a stronger claim.

The claim that survives is a system-level one: a North-Indian-Ocean-focused, independently-validated reconstruction system - surface to subsurface temperature with uncertainty, anomaly and an observation-priority layer - built and verified end to end. We are not proposing a new method, and we do not need to be.

From the project's own novelty matrix, which corrected its own earlier assessment.

IN ONE LINE

SIH26066 asks whether the surface picture satellites already give us can be turned into the depth-resolved picture only sparse floats can currently provide. OceanEmbed answers yes, at 0.9006 degC against 962 independent floats - and refuses to quote the more flattering number that came from the wrong inputs.

16 / 0 / 1

PS REQUIREMENTS: PASS / FAIL / BLOCKED

24,000 x 15

CELLS X DEPTHS, EVERY DAY

962

INDEPENDENT PROFILES BEHIND THE HEADLINE

Note 1 of 19. Next: Feature 1 - the Phase-1 App.